

Generation of cell polarity in plants links endocytosis, auxin distribution and cell fate decisions

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Evidence abstract

Dynamically polarized membrane proteins define different cell boundaries and have an important role in intercellular communication-a vital feature of multicellular development.

Efflux carriers for the signalling molecule auxin from the PIN family are landmarks of cell polarity in plants and have a crucial involvement in auxin distribution-dependent development including embryo patterning, organogenesis and tropisms.

Polar PIN localization determines the direction of intercellular auxin flow, yet the mechanisms generating PIN polarity remain unclear.

Here we identify an endocytosis-dependent mechanism of PIN polarity generation and analyse its developmental implications.

Real-time PIN tracking showed that after synthesis, PINs are initially delivered to the plasma membrane in a non-polar manner and their polarity is established by subsequent endocytic recycling.

Interference with PIN endocytosis either by auxin or by manipulation of the Arabidopsis Rab5 GTPase pathway prevents PIN polarization.

Failure of PIN polarization transiently alters asymmetric auxin distribution during embryogenesis and increases the local auxin response in apical embryo regions.

This results in ectopic expression of auxin pathway-associated root-forming master regulators in embryonic leaves and promotes homeotic transformation of leaves to roots.

Our results indicate a two-step mechanism for the generation of PIN polar localization and the essential role of endocytosis in this process.

It also highlights the link between endocytosis-dependent polarity of individual cells and auxin distribution-dependent cell fate establishment for multicellular patterning.

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Evidence checklist

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2. Efflux carriers for the signalling molecule auxin from the PIN family are landmarks of cell polarity in plants and have a crucial involvement in auxin distribution-dependent development including embryo patterning, organogenesis and tropisms.
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embryogenesis and increases the local auxin response in apical embryo regions.

8. This results in ectopic expression of auxin pathway-associated root-forming master regulators in embryonic leaves and promotes homeotic transformation of leaves to roots.

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Source continuity

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